

Land Survey

Engineering Mathematics II

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\text{Gradient} = \frac{\text{Rise}}{\text{Run}}$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\Delta E = L \sin \theta$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\Delta N = L \cos \theta$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$A = \frac{1}{2} \left| \sum x_i y_{i+1} - \sum y_i x_{i+1} \right|$$

Section B

5. Calculate reduced levels from levelling data. **(10 marks)**
6. Compute latitude and departure. **(10 marks)**
7. Find area from coordinates. **(10 marks)**
8. Create one realistic land survey problem based on this topic and solve it fully. **(20 marks)**